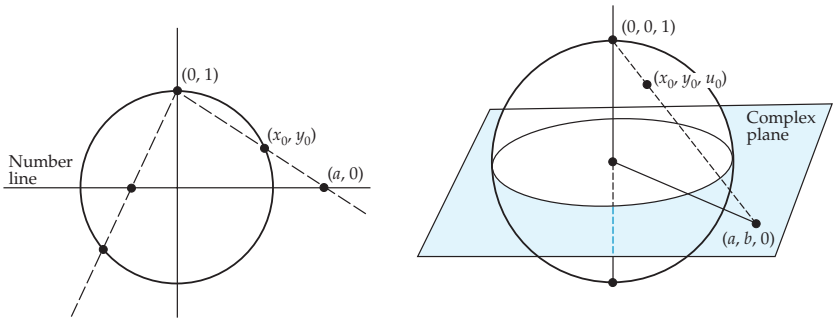
allow $|z|$ to become unbounded, say, along the real and imaginary axes, we do not have to distinguish between "directions" on these axes by notations such as $z \to +\infty$, $z \to -\infty$, $z \to +i\infty$, and $z \to -i\infty$. In complex analysis, only a single notion of ∞ is used because we can extend the complex number system **C** in a manner analogous to that just described for the real number system **R**. This time, however, we associate a complex number with a point on a unit sphere called the **Riemann sphere**. By drawing a line from the number $z = a + ib$, written as $(a,\ b,\ 0)$, in the complex plane to the north pole $(0,\ 0,\ 1)$ of the sphere $x^2 + y^2 + u^2 = 1$, we determine a unique point $(x_0,\ y_0,\ u_0)$ on a unit sphere. As can be visualized from Figure 1.5.10(b), a complex number with a very large modulus is far from the origin (0, 0, 0) and, correspondingly, the point $(x_0,\ y_0,\ u_0)$ is close to $(0,\ 0,\ 1)$. In this manner each complex number is identified with a single point on the sphere. See Problems 48–50 in Exercises 1.5. Because the point $(0,\ 0,\ 1)$ corresponds to no number z in the plane, we correspond it with ∞. Of course, the system consisting of **C** adjoined with the "ideal point" ∞ is called the **extended complex-number system**.

This way of corresponding or mapping the complex numbers onto a sphere—north pole $(0,\ 0,\ 1)$ excluded—is called a **stereographic projection**.

For a finite number z, we have $z + \infty = \infty + z = \infty$, and for $z \neq 0$, $z \cdot \infty = \infty \cdot z = \infty$. Moreover, for $z \neq 0$ we write $z/0 = \infty$ and for $z \neq \infty$, $z/\infty = 0$. Expressions such as $\infty - \infty$, ∞/∞, ∞^0, and 1^∞ cannot be given a meaningful definition and are called indeterminate.



(a) Unit circle **(b)** Unit sphere

FIGURE 1.5.10 The method of correspondence in (b) is a stereographic projection

EXERCISES 1.5

Answers to selected odd-numbered problems begin on page ANS-3.

In Problems 1–12, sketch the graph of the given equation in the complex plane.

1. $|z - 4 + 3i| = 5$

2. $|z + 2 + 2i| = 2$

3. $|z + 3i| = 2$

4. $|2z - 1| = 4$

5. $\text{Re}(z) = 5$

6. $\text{Im}(z) = -2$

7. $\text{Im}(\bar{z} + 3i) = 6$

8. $\text{Im}(z - i) = \text{Re}(z + 4 - 3i)$

9. $|\text{Re}\,(1 + i\bar{z})| = 3$

10. $z^2 + \bar{z}^2 = 2$

11. $\text{Re}(z^2) = 1$

12. $\arg(z) = \pi/4$

In Problems 13–24, sketch the set S of points in the complex plane satisfying the given inequality. Determine whether the set is (**a**) open, (**b**) closed, (**c**) a domain, (**d**) bounded, or (**e**) connected.

13. $\text{Re}(z) < -1$

14. $|\text{Re}\,(z)| > 2$

15. $\text{Im}(z) > 3$

16. $\text{Re}\,((2+i)z+1) > 0$

17. $2 < \text{Re}(z-1) < 4$

18. $-1 \le \text{Im}(z) < 4$

19. $\text{Re}(z^2) > 0$

20. $\text{Im}(z) < \text{Re}(z)$

21. $|z-i| > 1$

22. $2 < |z-i| < 3$

23. $1 \le |z-1-i| < 2$

24. $2 \le |z-3+4i| \le 5$

25. Give the boundary points of the sets in Problems 13–24.

26. Consider the set S consisting of the complex plane with the circle $|z| = 5$ deleted. Give the boundary points of S. Is S connected?

In Problems 27 and 28, sketch the set of points in the complex plane satisfying the given inequality.

27. $0 \le \arg(z) \le \pi/6$

28. $-\pi < \arg(z) < \pi/2$

In Problems 29 and 30, describe the shaded set in the given figure using $\arg(z)$ and an inequality.

29.

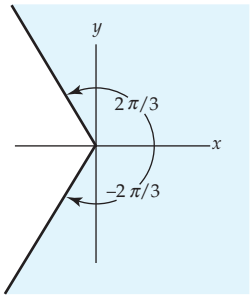


FIGURE 1.5.11 Figure for Problem 29

30.

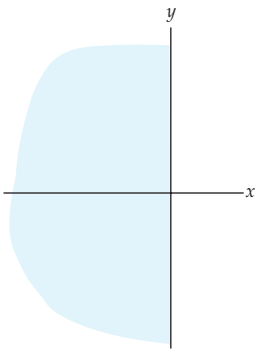


FIGURE 1.5.12 Figure for Problem 30

In Problems 31 and 32, solve the given pair of simultaneous equations.

31. $|z| = 2, \quad |z-2| = 2$

32. $|z-i| = 5, \quad \arg(z) = \pi/4$

Focus on Concepts

33. On page 29 we stated that if $\rho_1 > 0$, then the set of points satisfying $\rho_1 < |z - z_0|$ is the exterior to the circle of radius ρ_1 centered at z_0. In general, describe the set if $\rho_1 = 0$. In particular, describe the set defined by $|z + 2 - 5i| > 0$.

34. (**a**) What are the boundary points of a deleted neighborhood of z_0?

(**b**) What are the boundary points of the complex plane?

(**c**) Give several examples, not including the one given on page 30, of a set S in the complex plane that is neither open nor closed.

35. Use complex notation and inequalities in parts (a) and (b).

(**a**) Make up a list of five sets in the complex plane that are connected.

(**b**) Make up a list of five sets in the complex plane that are not connected.

36. Consider the disk centered at z_0 defined by $|z - z_0| \leq \rho$. Demonstrate that this set is bounded by finding an $R > 0$ so that all points z in the disk satisfy $|z| < R$. [*Hint*: See the discussion on inequalities in Section 1.2.]

37. Suppose z_0 and z_1 are distinct points. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfy $|z - z_0| = |z - z_1|$.

38. Using only the concept of distance, describe in words the set of points z in the complex plane that satisfies $|z - i| + |z + i| = 1$.

In Problems 39 and 40, describe the shaded set in the given figure by filling in the two blanks in the set notation

$$\{\, z\text{: } _____ \text{ and/or } _____ \}$$

using complex notation for equations or inequalities and one of the words *and* or *or*.

39.

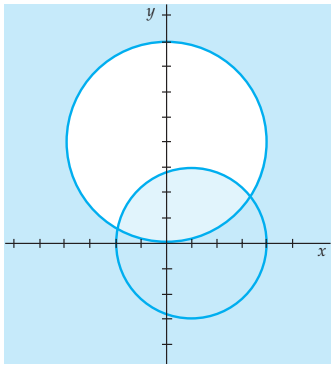


FIGURE 1.5.13 Figure for Problem 39

40.

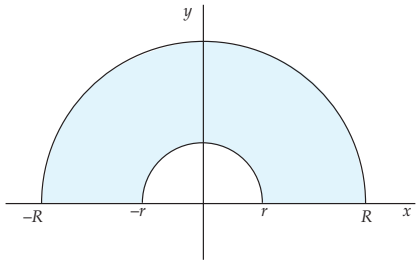


FIGURE 1.5.14 Figure for Problem 40

41. Consider the set S of points in the complex plane defined by $\{i/n\}$, $n = 1, 2, 3, \ldots$. Discuss which of the following terms apply to S: *boundary*, *open*, *closed*, *connected*, *bounded*.

42. Consider a finite set S of complex numbers $\{z_1, z_2, z_3, \ldots, z_n\}$. Discuss whether S is necessarily bounded. Defend your answer with sound mathematics.

43. A set S is said to be **convex** if each pair of points P and Q in S can be joined by a line segment $\overline{PQ}$ such that every point on the line segment also lies in S. Determine which of the sets S in the complex plane defined by the following conditions are convex.

(a) $|z - 2 + i| < 3$

(b) $1 < |z| < 2$

(c) $x > 2,\ y \leq -1$

(d) $y < x^2$

(e) $\text{Re}(z) \leq 5$

(f) $\text{Re}(z) \neq 0$

44. Discuss: Is a convex set, defined in Problem 43, necessarily connected?

45. Discuss and decide whether the empty set $\varnothing$ is an open set.

46. Suppose S_1 and S_2 are open sets in the complex plane.

(a) Discuss: Is the union $S_1 \cup S_2$ an open set? If you think the statement is true, try to prove it. If you think the statement is false, find a counterexample.

(b) Repeat part (a) for the intersection $S_1 \cap S_2$.

Before answering Problems 47–50, reread the *Remarks* on the end of this section.

47. Find the point $(x_0,\ y_0)$ on the unit circle that corresponds to each of the real numbers: $-\frac{1}{4},\ \frac{1}{2},\ -3,\ 1,\ 10$. See Figure 1.5.10(a).

48. Find a point $(x_0,\ y_0,\ u_0)$ on the unit sphere that corresponds to the complex number $2 + 5i$. See Figure 1.5.10(b).

49. Describe the set of points on the unit sphere that correspond to each of the following sets in the complex plane.

(**a**) the numbers on unit circle $|z| = 1$

(**b**) the numbers within the open disk $|z| < 1$

(**c**) the numbers that are exterior to unit circle, that is, $|z| > 1$

50. Express the coordinates of the point $(x_0,\ y_0,\ u_0)$ on the unit sphere in Figure 1.5.10(b) in terms of the coordinates of the point $(a,\ b,\ 0)$ in the complex plane. Use these formulas to verify your answer to Problem 48. [*Hint*: First show that all points on the line containing $(0,\ 0,\ 1)$ and $(a,\ b,\ 0)$ are of the form $(ta,\ tb, 1 - t)$.]

1.6 Applications

INTRODUCTION In this section we are going to examine a few simple applications of complex numbers. It will be assumed in the discussion that the reader has some familiarity with methods for solving elementary ordinary differential equations.

We saw how to find roots of complex numbers in Section 1.4. With that background we are now in a position to examine how to solve a quadratic equation with complex coefficients using the complex version of the quadratic formula. We then examine how complex numbers and the complex exponential are used in differential equations. This last discussion leads us to Euler's formula and a new compact way of writing the polar form of a complex number. Lastly, we explore some ways complex numbers are used in electrical engineering.

Algebra

You probably encountered complex numbers for the first time in a beginning course in algebra where you learned that roots of polynomial equations can be complex as well as real. For example, any second degree, or quadratic, polynomial equation can be solved by completing the square. In the general case, $ax^2+bx+c = 0$, where the coefficients $a \neq 0,\ b$, and c are real, completion of the square in x yields the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{1}$$

When the discriminant $b^2 - 4ac$ is negative, the roots of the equation are complex. For example, by (1) the two roots of $x^2 - 2x + 10 = 0$ are

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(10)}}{2(1)} = \frac{2 \pm \sqrt{-36}}{2}. \tag{2}$$

In beginning courses the imaginary unit i is written $i = \sqrt{-1}$ and the assumption is made that the laws of exponents hold so that a number such as $\sqrt{-36}$ can be written $\sqrt{-36} = \sqrt{36}\sqrt{-1} = 6i$. Let us denote the two complex roots in (2) as $z_1 = 1 + 3i$ and $z_2 = 1 - 3i$.

Note: The roots z_1 and z_2 are conjugates. See Problem 22 in Exercises 1.6. ▶

Quadratic Formula

The quadratic formula is perfectly valid when the coefficients $a \neq 0,\ b$, and c of a quadratic polynomial equation $az^2 + bz + c = 0$ are complex numbers. Although the formula can be obtained in exactly the same manner